

# Who Reports Withdrawal Dates in the Western Interconnection Queue, and Does It Matter?

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Draft, September 2026

## Abstract

Berkeley Lab’s *Queued Up* publishes a median duration from interconnection request to withdrawal for the non-ISO West. That figure is computed from the 12.1% of western withdrawals carrying a recorded withdrawal date: 637 of 5277. This paper asks whether those records represent the region they describe. They do not—technology mix, queue cohort and interconnection service type all differ from the regional population at  $p < 10^{-11}$ , while project capacity does not. But standardising the subset to the region’s actual composition moves the median by less than three months, with an interval including zero and a sign that reverses under a truncation-matched observation window. A validation exercise inside the well-covered ISO regions explains why: imposing the West’s compositional distortion on a population whose durations are fully observed shifts the median by about eight days. Technology, cohort and capacity together explain 4.6% of the variance in withdrawal duration, leaving coverage rather than composition as the binding caveat. A companion logistic regression over all 8097 western requests, requiring no dates, finds solar the most withdrawal-prone technology, larger projects likelier to be withdrawn, and roughly a quarter of the latent variation in withdrawal risk sitting between providers rather than between projects.

## Summary

Berkeley Lab’s *Queued Up* series publishes, among much else, a median duration from interconnection request to withdrawal for each region of the United States. For the non-ISO West that figure is computed from the withdrawals that carry a recorded withdrawal date. In the 2026 Edition data file, that is 637 of 5277 western withdrawals—12.1%. The remaining 87.9% are known to have been withdrawn but not when.

This paper asks whether those 637 records are representative of the region whose duration they are used to describe, and if not, how far the published median is displaced as a result. The answer has three parts.

**First, the reporting is extraordinarily concentrated.** Of the 31 western transmission providers with at least one withdrawal in the file, six record any withdrawal date at all and five record one for more than a quarter of their withdrawals. PacifiCorp—which alone accounts for 2160 of the region’s 8097 requests and 1627 of its withdrawals, more than twice the next-largest provider on either count—records a date for 6.0% of them. The median western withdrawal duration is, in practice, a statistic about Colorado, New Mexico, eastern Washington, Los Angeles and El Paso.

**Second, that subset is compositionally unlike the region.** Technology mix, queue cohort and interconnection service type all differ from the regional population at  $p < 10^{-11}$  after correction for multiple comparisons, on both of the two defensible definitions of “the reporting subset.” Project capacity does not differ. The dated records over-represent stand-alone solar

(44.8 % against 31.5 %) and wind (30.7 % against 23.1 %) and under-represent solar-plus-storage (10.7 % against 18.8 %) and stand-alone storage (3.6 % against 9.0 %).

**Third—and this is the part that resists the obvious conclusion—those compositional differences barely move the median.** Standardising the reporting subset to the region’s actual technology-by-cohort composition shifts the median by  $-0.06$  years, with a 95 % bootstrap interval of  $[-0.23, +0.00]$ . Under a truncation-matched window the shift is  $+0.19$  years. The sign flips. A validation exercise inside the ISO regions, where date coverage is near-complete and the answer is therefore known, shows why: deliberately imposing the West’s skewed composition on those regions biases their median by 0.021 years—about eight days. Technology, cohort and capacity together explain 4.6 % of the variance in withdrawal duration where it can be measured well. Composition is simply not a strong enough determinant of duration for a composition gap of this size to matter much.

The correct caveat on the published western median is therefore not “it is biased by an unrepresentative technology mix.” It is “it is computed from one eighth of the region’s withdrawals, drawn from five of thirty-one providers, and the mechanism by which that might bias it is unmeasured.” Composition is the part we can check, and it is small. What we cannot check is whether providers that record dates withdraw on a different *schedule* from providers that do not—and nothing in this data file can settle that.

## 1 The question

Interconnection queues have become the binding constraint on adding generation to the North American grid, and *Queued Up* is the reference description of them. Its regional summaries are widely cited in regulatory filings, trade press and academic work. One of those summaries is the typical time a request spends in the queue before being withdrawn—a natural measure of how much developer effort the process absorbs before yielding nothing.

Computing that statistic requires two dates: when the request entered the queue, and when it was withdrawn. The first is recorded almost universally. The second is not. Transmission providers publish their queues in whatever format they choose, and many simply drop a withdrawn request from the published list without recording the date on which it left. The compilers of *Queued Up* can only report what providers publish.

Where coverage is near-complete this is a non-issue. In the four regions with the best coverage—ISO-NE, CAISO, PJM and MISO—94 % to 99 % of withdrawals carry a date, so the observed durations are effectively a census. In the non-ISO West they are 12.1 % of withdrawals, and the question of who those 12.1 % are becomes unavoidable.

Two things could go wrong. The reporting subset could differ from the region in project characteristics that themselves influence how long a request survives—a subset heavy in wind, say, when wind takes longer to withdraw than solar. Call this **compositional bias**; it is measurable, because both the composition of the subset and the composition of the region are observed. Or the providers that record dates could simply process requests differently from those that do not, in ways not captured by any recorded project attribute. Call this **provider process bias**; it is not measurable from this file, because the durations of the non-reporting providers are exactly what is missing.

This paper measures the first and bounds what can be said about the second.

## 2 Data

The analysis uses a single source: the *Queued Up* 2026 Edition data file (LBNL\_Ix\_Queue\_Data\_File\_thru2025.x covering interconnection requests through the end of 2025, published by Lawrence Berkeley National Laboratory and GridTracker under CC BY 4.0. Sheet 03. **Complete Queue Data** holds project-level records: 38,201 rows across nine regions.

The **region** field distinguishes the seven ISO/RTO territories, the Southeast, and “West”—the non-ISO Western Interconnection, i.e. WECC excluding CAISO. That western region is the study population: 8097 **requests**, of which 5277 (65.2 %) are recorded as withdrawn, 1603 active, 963 operational, 244 suspended and 10 unknown.

Four fields carry the analysis. **q\_date** and **wd\_date** give the request and withdrawal dates; their difference, in years, is the duration. **type\_clean** gives a cleaned technology label, which I collapse to six groups (Solar, Wind, Solar+Battery, Battery, Gas, Other). **q\_year** gives the queue entry year, binned into four cohorts ( $\leq 2014$ , 2015–18, 2019–21, 2022+). **mw\_1** gives nameplate capacity, and **entity** the transmission provider.

A record is **dated** if it carries both a queue date and a withdrawal date and the resulting duration is non-negative. Six records across the whole file have withdrawal dates preceding their queue dates and are excluded on that ground. By this definition 637 of the West’s 5277 withdrawals are dated; one of those lacks a queue year and so drops out of the cohort-stratified analysis, leaving 636.

Two definitions of “the reporting subset” are carried in parallel throughout, because they are not the same set and the choice turns out to matter more than any of the corrections applied to them:

- **dated records**—the 636 withdrawn requests carrying a usable duration. This is what the published median literally rests on, irrespective of who filed them.
- **reporting entities**—the 539 dated withdrawals filed with the five providers that record a date for at least a quarter of their withdrawals. This is the institutional unit: the set of providers whose reporting practice makes the statistic possible.

The gap between them is exactly 97 dated records, every one of them PacifiCorp’s.

**Table 1:** Withdrawal-date coverage by region.

| Region      | Withdrawals | Dated      | Coverage |
|-------------|-------------|------------|----------|
| ISO-NE      | 982         | 970        | 98.8 %   |
| CAISO       | 2200        | 2160       | 98.2 %   |
| PJM         | 5226        | 5091       | 97.4 %   |
| MISO        | 3263        | 3060       | 93.8 %   |
| SPP         | 1846        | 1340       | 72.6 %   |
| NYISO       | 1531        | 745        | 48.7 %   |
| ERCOT       | 1158        | 520        | 44.9 %   |
| Southeast   | 2738        | 845        | 30.9 %   |
| <b>West</b> | 5277        | <b>637</b> | 12.1 %   |

The West is not merely the worst-covered region; it is worse by a factor of two and a half than the next-worst. This ordering also defines the calibration set used in Section 5: the four regions above 90 % coverage.

### 3 The roster

The concentration of reporting is the single most striking feature of the western data, and it is worth setting out in full before any modelling.

Thirty-six entities appear in the western region. Five have no withdrawals at all. Of the remaining 31, **six record at least one withdrawal date, and five record one for at least a quarter of their withdrawals.**

**Table 2:** Western transmission providers with 25 or more requests, through 2025. *Dated* counts withdrawals carrying a usable duration; *Rate* is that as a share of the provider’s withdrawals. Reporting providers in bold.

| Provider      | Requests | Withdrawn | Dated | Rate   |
|---------------|----------|-----------|-------|--------|
| PacifiCorp    | 2160     | 1627      | 97    | 6.0 %  |
| BPA           | 998      | 611       | 0     | 0 %    |
| Idaho Power   | 824      | 515       | 0     | 0 %    |
| APS           | 619      | 426       | 0     | 0 %    |
| NV Energy     | 505      | 263       | 0     | 0 %    |
| NorthWestern  | 448      | 328       | 0     | 0 %    |
| <b>PSCo</b>   | 408      | 295       | 286   | 96.9 % |
| <b>PNM</b>    | 356      | 224       | 133   | 59.4 % |
| WAPA-IS       | 273      | 194       | 0     | 0 %    |
| <b>Avista</b> | 173      | 117       | 71    | 60.7 % |
| SRP           | 155      | 105       | 0     | 0 %    |
| PSE           | 142      | 86        | 0     | 0 %    |
| TEP           | 120      | 84        | 0     | 0 %    |
| PGE           | 114      | 51        | 0     | 0 %    |
| <b>LADWP</b>  | 113      | 55        | 36    | 65.5 % |
| WAPA-RM       | 110      | 88        | 0     | 0 %    |
| TSGT          | 104      | 14        | 0     | 0 %    |
| <b>EPE</b>    | 66       | 30        | 14    | 46.7 % |
| PRPA          | 62       | 35        | 0     | 0 %    |
| BHCT          | 51       | 32        | 0     | 0 %    |
| SRP-ANPP      | 45       | 18        | 0     | 0 %    |
| CLPT          | 40       | 13        | 0     | 0 %    |
| WAPA-DSW      | 34       | 0         | 0     | —      |
| IID           | 33       | 0         | 0     | —      |
| SMUD          | 33       | 18        | 0     | 0 %    |
| SRP-PV/PC     | 33       | 22        | 0     | 0 %    |
| BHP           | 26       | 7         | 0     | 0 %    |

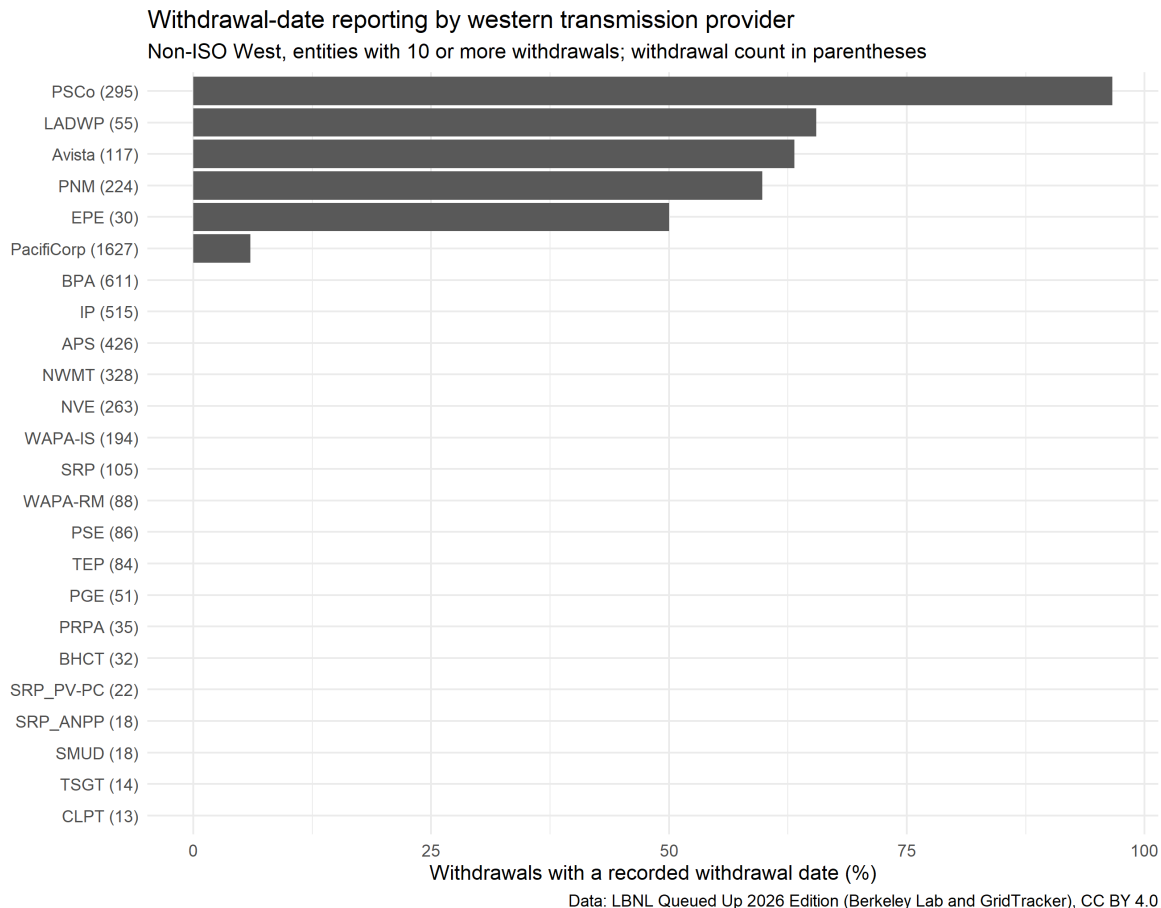
The remaining nine providers have fewer than 25 requests each and none records a date.

The distribution is close to binary. Providers either record dates for roughly half their withdrawals or better, or they record none whatsoever. PacifiCorp is the sole exception, and its 6.0 % is the reason the count of reporting providers reads six or five depending on where the line is drawn—a distinction with real consequences, since PacifiCorp is by a wide margin the largest queue in the region.

That PacifiCorp is both the largest western queue and a near-total non-reporter is the central fact of this paper. Its 1627 withdrawals are 31 % of the region’s total, and 2.7 times those of the next-largest provider. Whatever the typical PacifiCorp withdrawal duration is, it contributes

6 % of its weight to the published median.

Note also the geographic consequence. The five reporting providers serve Colorado (PSCo), New Mexico and west Texas (PNM, EPE), eastern Washington and northern Idaho (Avista), and the City of Los Angeles (LADWP). The published western median describes those service territories. It is nearly silent on Oregon, Montana, Wyoming, Nevada, Arizona and the Bonneville footprint. I do not report a formal test of geographic difference, because provider and state are so nearly collinear in this region that such a test would only restate Table 2.



**Figure 1:** Withdrawal-date reporting rate by western transmission provider, for providers with ten or more withdrawals.

## 4 Is the reporting subset different?

### 4.1 Method

I test five pre-specified comparisons between the reporting subset and the rest of the region. Four are project attributes that the ISO calibration in Section 5 shows to influence duration; the fifth is a provider-level property.

1. **Technology mix**—six groups, Pearson  $\chi^2$ .
2. **Queue cohort mix**—four bins, Pearson  $\chi^2$ .
3. **Capacity**—mw\_1, Wilcoxon rank-sum.

4. **Interconnection service type**—ERIS / NRIS / NRIS-ERIS / Other,  $\chi^2$ .
5. **Withdrawal rate**—the share of all requests ending in withdrawal,  $\chi^2$ , computed over all 8097 requests rather than the withdrawals alone.

Each family of five is Holm-corrected within basis. Both bases receive the full family, giving ten tests in all. For test 5 each basis takes its natural provider-level analogue: for the dated basis, providers recording any date (six, including PacifiCorp); for the entity basis, providers clearing the 25 % threshold (five).

## 4.2 Results

**Table 3:** The family of five comparisons, Holm-adjusted within basis.

| Basis              | Comparison       | Statistic         | $p$ (Holm)            | Verdict           |
|--------------------|------------------|-------------------|-----------------------|-------------------|
| dated records      | technology mix   | $\chi^2 = 129.04$ | $1.9 \times 10^{-25}$ | <b>difference</b> |
| dated records      | queue cohort mix | $\chi^2 = 70.70$  | $9.1 \times 10^{-15}$ | <b>difference</b> |
| dated records      | capacity (MW)    | $W = 1,466,743$   | 0.80                  | null              |
| dated records      | service type     | $\chi^2 = 54.19$  | $2.0 \times 10^{-11}$ | <b>difference</b> |
| dated records      | withdrawal rate  | $\chi^2 = 101.95$ | $2.3 \times 10^{-23}$ | <b>difference</b> |
| reporting entities | technology mix   | $\chi^2 = 120.19$ | $1.1 \times 10^{-23}$ | <b>difference</b> |
| reporting entities | queue cohort mix | $\chi^2 = 142.76$ | $4.8 \times 10^{-30}$ | <b>difference</b> |
| reporting entities | capacity (MW)    | $W = 1,562,008$   | 0.068                 | null              |
| reporting entities | service type     | $\chi^2 = 58.54$  | $3.6 \times 10^{-12}$ | <b>difference</b> |
| reporting entities | withdrawal rate  | $\chi^2 = 0.16$   | 0.69                  | null              |

Three comparisons return a difference on both bases: **technology mix**, **queue cohort mix** and **interconnection service type**. One returns a null on both: **capacity**. One splits: **withdrawal rate**.

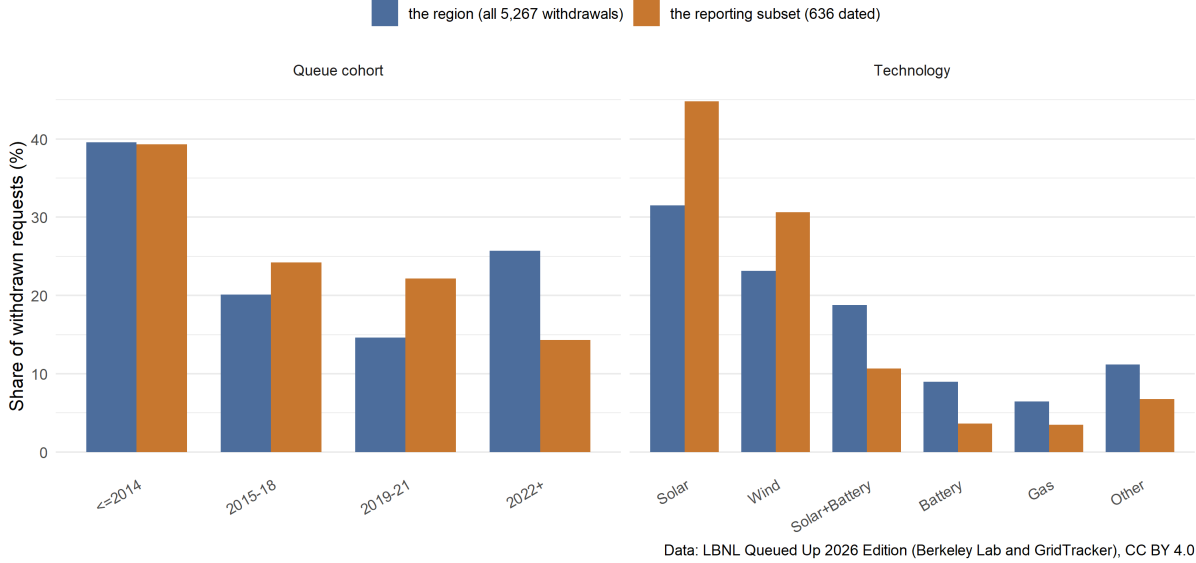
The split is instructive rather than awkward. Providers recording any date withdraw 71.7 % of their requests against 60.8 % for the rest—but that gap is entirely PacifiCorp, whose own withdrawal rate is 75.3 % and whose 2160 requests dominate the group. Restrict to the five genuine reporters and the rates are 64.6 % against 65.3 %, a difference of seven tenths of a percentage point ( $p = 0.69$ ). Providers that actually report dates do not withdraw at an unusual rate. The apparent effect on the wider definition is one large utility’s idiosyncrasy.

Against the regional population of 5267 withdrawals, the 636 dated records over-represent stand-alone solar (44.8 % against 31.5 %) and wind (30.7 % against 23.1 %), and under-represent solar-plus-storage (10.7 % against 18.8 %), stand-alone storage (3.6 % against 9.0 %), gas (3.5 % against 6.5 %) and other technologies (6.8 % against 11.2 %). On cohort they are close on the oldest bin (39.3 % against 39.6 %) but over-represent 2019–21 (22.2 % against 14.6 %) and under-represent 2022 and later (14.3 % against 25.7 %).

The shape of this is coherent. Storage and hybrid projects entered western queues in volume from about 2019, and did so disproportionately in the territories of the large non-reporting providers. The reporting subset is therefore both older and more conventional in technology than the region it stands for.

The withdrawals that carry a date are not the withdrawals the region files

Non-ISO West, requests through 2025. Technology and cohort mix both differ at  $p < 1e-14$ ; capacity does not differ.



**Figure 2:** Technology and queue-cohort composition of the reporting subset against the region it represents.

## 5 How far does the composition gap move the median?

Establishing that the reporting subset is unrepresentative does not establish that the published median is wrong. That requires knowing how strongly the attributes that differ actually influence duration—and then reweighting.

The estimation proceeds in three stages: calibrate the influence of technology and cohort on duration where coverage is good; validate the reweighting estimator in that same place, where the true answer is known; then apply it to the West.

### 5.1 Stage A: what moves duration, measured where we can see it

The four regions above 90 % coverage—CAISO, ISO-NE, MISO, PJM—supply 11,281 dated withdrawals whose durations are effectively a census rather than a sample. Regressing  $\log(\text{duration} + 1/12)$  on technology, cohort and log capacity there gives the effects in Table 4.

**Table 4:** ISO calibration ( $n = 11,281$ ). Multiplicative effect on duration.

| Term                             | Multiplier | $p$                   |
|----------------------------------|------------|-----------------------|
| Wind (vs Solar)                  | 1.447      | $1.5 \times 10^{-34}$ |
| Solar+Battery                    | 1.144      | $7.1 \times 10^{-5}$  |
| Other                            | 1.076      | 0.043                 |
| Battery                          | 1.031      | 0.27                  |
| Gas                              | 1.005      | 0.88                  |
| cohort 2015–18 (vs $\leq 2014$ ) | 0.849      | $2.4 \times 10^{-9}$  |
| cohort 2019–21                   | 1.420      | $6.1 \times 10^{-37}$ |
| cohort 2022+                     | 1.102      | 0.0022                |
| log capacity (MW)                | 1.016      | 0.012                 |

Wind withdrawals take about 45 % longer than solar; the 2019–21 cohort—the queue surge that has been working through withdrawal since roughly 2024—takes about 42 % longer than the pre-2015 cohorts. These are precisely estimated and substantively real.

**But the model’s  $R^2$  is 0.046.** Technology, cohort and size together account for under 5 % of the variance in how long a request takes to be withdrawn, in the data where we can measure that variance cleanly. This single number governs everything that follows: composition can differ a great deal without the median moving much, because composition is a weak determinant of duration.

## 5.2 Stage B: does the correction work where the answer is known?

Before applying a reweighting estimator to the West, it is worth checking that it does what it claims on data where the truth is observable.

The true median duration across the four calibration regions is 1.629 years. I repeatedly draw subsamples of roughly 600 records from those regions, constructed to match the *West’s dated* technology-by-cohort composition—that is, deliberately distorted in exactly the way the western reporting subset is distorted. I then ask whether the reweighting recovers 1.629 from the distorted subsample.

**Table 5:** Validation inside the calibration regions, 500 replicates.

| Estimator                        | Mean  | Bias   | RMSE  |
|----------------------------------|-------|--------|-------|
| distorted subsample, uncorrected | 1.650 | +0.021 | 0.051 |
| + post-stratification            | 1.633 | +0.004 | 0.052 |
| + model-assisted transport       | 1.674 | +0.045 | 0.092 |

Two conclusions. First, post-stratification works: it removes essentially all of the compositional bias, cutting it from 0.021 to 0.004 years. Second, and far more important, **the bias it removes is tiny to begin with.** Imposing the West’s composition on a population of 11,281 well-measured durations shifts the median by 0.021 years—about eight days.

The model-assisted transport estimator, which strips out the calibrated cell offsets and re-attaches them under the target composition, performs *worse* than doing nothing. It is reported below for completeness but the validation disqualifies it as the preferred estimator; post-stratification is the one to read.

This result is the most consequential in the paper, and it constrains the conclusion before the western estimate is even computed. Whatever the western reporting problem is, a composition gap of this magnitude cannot produce a large displacement in the median.

## 5.3 Stage C: the western estimate

Strata are technology (6)  $\times$  cohort (4) = 24 cells. Cell coverage is good: only two cells contain no dated western record, and together they carry 0.08 % of the target weight; eight cells hold fewer than five dated records and carry 5.3 %. Post-stratification weights each observed duration by its cell’s share of the 5267-withdrawal regional population divided by the number of dated records in that cell, renormalising over covered cells. Confidence intervals come from 2000 bootstrap replicates of the shift.

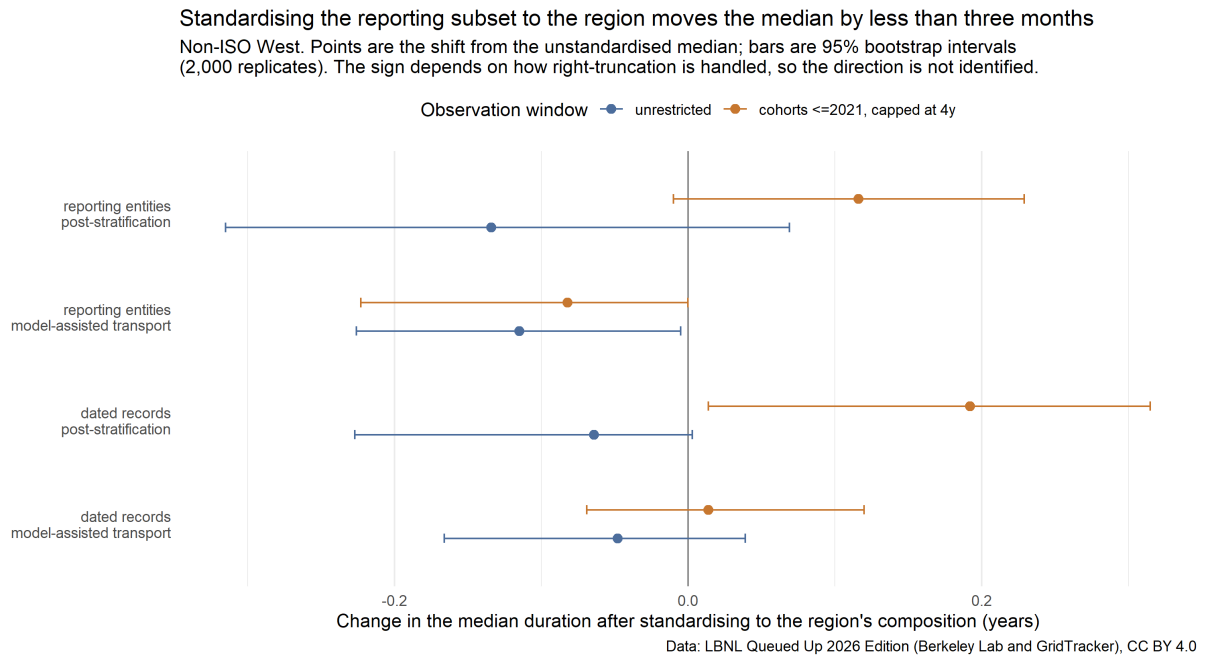
Read the post-stratification rows. On the unrestricted window the median falls from 1.83 to 1.77 years—a shift of about three weeks, with an interval that includes zero. On the matched

**Table 6:** Standardised medians, in years. “Matched” restricts to cohorts entering in 2021 or earlier and caps durations at four years.

| Window       | Basis              | Estimator   | Unstd. | Std.  | Shift         | 95 % CI          |
|--------------|--------------------|-------------|--------|-------|---------------|------------------|
| unrestricted | dated records      | post-strat. | 1.830  | 1.766 | <b>−0.064</b> | [−0.227, +0.003] |
| unrestricted | dated records      | transport   | 1.830  | 1.782 | −0.048        | [−0.166, +0.039] |
| unrestricted | reporting entities | post-strat. | 1.552  | 1.418 | <b>−0.134</b> | [−0.315, +0.069] |
| unrestricted | reporting entities | transport   | 1.552  | 1.437 | −0.115        | [−0.226, −0.005] |
| matched      | dated records      | post-strat. | 2.136  | 2.327 | <b>+0.192</b> | [+0.014, +0.315] |
| matched      | dated records      | transport   | 2.136  | 2.150 | +0.014        | [−0.069, +0.120] |
| matched      | reporting entities | post-strat. | 1.808  | 1.925 | <b>+0.116</b> | [−0.010, +0.229] |
| matched      | reporting entities | transport   | 1.808  | 1.727 | −0.082        | [−0.223, 0.000]  |

window it rises from 2.14 to 2.33—about ten weeks, with an interval that excludes zero. The two windows disagree in sign.

Note also what dwarfs both corrections. Moving from the dated-records basis to the reporting-entities basis moves the unstandardised median from 1.83 to 1.55 years—0.28 years, four times the size of the unrestricted standardisation shift. **The definitional choice of which records count as “the reporting subset” moves the answer considerably more than correcting that subset’s composition does.** Any published duration statistic for this region carries that definitional sensitivity whether or not it acknowledges it.



**Figure 3:** Change in the median duration after standardising to the region’s composition, with 95 % bootstrap intervals.

## 5.4 Why the sign is not identified

Duration here is not a complete measurement. A request entering the queue in 2023 cannot display a five-year withdrawal duration, because five years have not elapsed. The observation window is bounded above by the file’s cutoff, and that bound bites harder on recent cohorts.

This interacts directly with the composition gap. The 2022+ cohort is 25.7 % of the regional

population but only 14.3 % of the dated subset. Reweighting therefore transfers a quarter of the weight onto the cohort whose durations are most severely truncated—which mechanically pulls the median down, regardless of any genuine difference in how long those projects take.

The matched window removes that mechanism: restricting to cohorts entering by 2021 gives every remaining record at least four years of exposure, and capping durations at four years makes the measurement identical across cohorts. Under that restriction the shift is positive.

Both estimates are defensible and they disagree. The unrestricted estimate targets exactly the quantity *Queued Up* publishes, and inherits its truncation. The matched estimate is a cleaner contrast but describes a different, more restricted quantity, discarding 26 % of the regional population to get there. Neither is wrong; they answer slightly different questions, and the answers have opposite signs.

The honest reading is therefore: **the compositional bias in the published western median is small—under three months on any specification, and under one month on the specification that matches the published estimand—and its direction is not identified by this data.** Given the validation result in Table 5, which puts the compositional bias at about eight days when measured against a known truth, the small magnitude is the robust finding and the sign is noise around it.

## 6 Withdrawal as a binary outcome

The duration analysis is confined to 12 % of western withdrawals. Withdrawal *itself* is recorded for every request, so a model of whether a request is withdrawn can use the entire region without needing a single date. This is a different question—what predicts withdrawal, not how long it takes—but it is answerable on the full population, and it speaks to how much of the variation in queue outcomes is attributable to the provider.

### 6.1 Specification

The outcome is withdrawal against everything else (active, operational, suspended, unknown). Predictors are technology (six groups, solar as reference), log capacity, cohort (four bins) and entity. Of 8097 requests, 7991 have complete predictors; the 106 exclusions all lack a queue year.

Entity requires care. Seven providers, covering 90 requests, show no variation in the outcome—five have no withdrawals at all, two have nothing but—and so cannot be assigned an identified coefficient in a fixed-effects logistic model. The primary specification therefore treats entity as a random intercept, which shrinks such providers toward the grand mean rather than dropping them, and retains all 7991 requests. A fixed-effects fit, with those seven providers dropped and providers under 25 requests pooled, is reported alongside; so are fits dropping the ambiguous suspended and unknown statuses, and restricting to cohorts with at least four years of exposure.

### 6.2 Results

Three findings.

**Solar is the most withdrawal-prone technology in the western queue.** Every other group carries lower odds of withdrawal, from wind and storage at roughly half solar’s odds to gas and the miscellaneous group at about a third. The coefficients are stable to every alternative specification: the fixed-effects fit reproduces them to within 0.01, and dropping suspended and unknown statuses changes nothing material.

**Larger projects are more likely to be withdrawn,** at 1.13 times the odds per unit of log

**Table 7:** Odds of withdrawal, mixed-effects logistic ( $n = 7991$ ). Entity variance 1.159 (SD 1.077); intraclass correlation **0.261**.

| Term                             | Odds ratio | 95 % CI        | $p$                   |
|----------------------------------|------------|----------------|-----------------------|
| Wind (vs Solar)                  | 0.547      | [0.458, 0.652] | $1.9 \times 10^{-11}$ |
| Solar+Battery                    | 0.578      | [0.487, 0.685] | $2.7 \times 10^{-10}$ |
| Battery                          | 0.521      | [0.428, 0.635] | $9.9 \times 10^{-11}$ |
| Gas                              | 0.355      | [0.282, 0.447] | $9.5 \times 10^{-19}$ |
| Other                            | 0.314      | [0.264, 0.373] | $4.6 \times 10^{-39}$ |
| log capacity (MW)                | 1.129      | [1.089, 1.170] | $3.2 \times 10^{-11}$ |
| cohort 2015–18 (vs $\leq 2014$ ) | 0.610      | [0.516, 0.721] | $7.6 \times 10^{-9}$  |
| cohort 2019–21                   | 0.446      | [0.374, 0.531] | $2.0 \times 10^{-19}$ |
| cohort 2022+                     | 0.344      | [0.293, 0.403] | $1.7 \times 10^{-39}$ |

capacity—roughly 9 % higher odds per doubling of nameplate capacity. This strengthens to 1.18 under the equal-exposure restriction.

**About a quarter of the latent variation in withdrawal risk sits between providers.**

The intraclass correlation of 0.261 says that a project’s odds of withdrawal depend substantially on whose queue it is in, after accounting for what kind of project it is, how big it is and when it entered. The provider random effect has a standard deviation of 1.077 on the log-odds scale, comparable to the largest technology contrast in the table—the miscellaneous group, at 1.158—and larger than every other.

That last result matters for the paper’s main question. It establishes that providers differ materially in queue outcomes even conditional on observable project attributes. It does not establish that they differ in withdrawal *timing*—the two are distinct, and timing is what the published median measures. But it removes any comfortable presumption that providers are interchangeable once technology and cohort are controlled for, which is exactly the presumption the compositional correction in Section 5 has to make.

### 6.3 The cohort coefficients are exposure, not behaviour

The declining odds across cohorts should not be read as a falling propensity to withdraw. They are dominated by exposure time.

**Table 8:** Withdrawal rate and exposure time by cohort.

| Cohort      | Requests | Withdrawn | Rate   | Median years of exposure |
|-------------|----------|-----------|--------|--------------------------|
| $\leq 2014$ | 2752     | 2084      | 75.7 % | 17                       |
| 2015–18     | 1457     | 1060      | 72.8 % | 9                        |
| 2019–21     | 1263     | 770       | 61.0 % | 6                        |
| 2022+       | 2519     | 1353      | 53.7 % | 3                        |

A request filed in 2024 has had two years in which to be withdrawn; one filed in 2010 has had fifteen. Restricting to cohorts with at least four years of exposure attenuates the cohort coefficients (2019–21 moves from 0.446 to 0.539) without touching the technology or capacity results, which is what one would expect if the cohort gradient is largely an artefact of the observation window and the others are not. The substantive findings survive; the cohort gradient should be read as censoring.

## 6.4 What carries the prediction

**Table 9:** Model comparison.

| Model                   | AIC    |
|-------------------------|--------|
| project attributes only | 9752.1 |
| entity only             | 9816.8 |
| both                    | 9378.6 |

Neither project attributes nor provider identity subsumes the other; both contribute (likelihood-ratio  $\chi^2 = 456.2$  on 9 df for adding attributes to an entity-only model). In-sample accuracy at a 0.5 threshold is 71.1 % against a 65.9 % base rate—a modest improvement, consistent with the general picture that recorded attributes explain a limited share of queue outcomes.

## 7 Limitations

**The provider process channel is unmeasured and unmeasurable here.** This is the binding limitation. Section 5 corrects for differences in *what kind of projects* the reporting providers handle. It cannot correct for differences in *how* they handle them. If PacifiCorp’s withdrawal process is systematically slower or faster than PSCo’s for reasons not encoded in technology, capacity, cohort or service type, that difference is invisible in this file, because PacifiCorp’s withdrawal dates do not exist. The finding that 26 % of latent withdrawal-risk variation is between providers makes this channel a live concern rather than a formality, and it is entirely plausible that it is larger than the compositional channel measured here. Establishing that would require withdrawal dates from at least one large non-reporting provider—obtainable, in principle, from FERC Form 715 filings or direct queue archives, but not from this data file.

**The calibration assumes transportability.** Stage A estimates how technology and cohort shift duration in CAISO, ISO-NE, MISO and PJM, and the validation is conducted within those same regions. Applying those effects to the West assumes the relationships carry over. Western interconnection processes differ from ISO tariff processes in ways that could plausibly alter them. This assumption is load-bearing only for the model-assisted transport estimator, which the validation already disqualifies; the preferred post-stratification estimator uses the ISO regions solely to establish that technology and cohort matter enough to be worth stratifying on, and takes its cell values from western data.

**Right-truncation is bounded, not eliminated.** The matched window addresses it, but capping at four years and discarding post-2021 cohorts is a blunt instrument. A survival-analytic treatment—modelling time to withdrawal with censoring for requests still active—would use the data more efficiently. It was not attempted here because the censoring pattern is itself entangled with the reporting problem: for non-reporting providers we know a request was withdrawn but not when, which is neither a standard observation nor a standard censoring event.

**Thin cells.** Eight of 24 standardisation cells hold fewer than five dated records, carrying 5.3 % of the target weight; two hold none, carrying 0.08 %. The bootstrap intervals reflect this, but a median computed from three observations is a fragile input regardless of how its uncertainty is propagated.

**Technology and cohort groupings are choices.** Six technology groups and four cohort bins were fixed before estimation on the criteria stated in Section 2—each cohort carrying at least 14 % of target weight, 2019–21 isolated as the surge, 2022+ isolated as the truncation-dominated bin—but finer or coarser schemes would give somewhat different numbers. Given the validation result, no plausible alternative stratification would change the magnitude conclusion.

**The outcome coding is a judgement.** Suspended (244 requests) and unknown (10) are treated as not withdrawn. Dropping them entirely leaves the coefficients essentially unchanged, so the choice is not consequential, but it is a choice.

**One data source, one vintage.** Everything rests on the 2026 Edition file. Reporting practice changes; several providers that record nothing today may begin to, and the composition of the reporting subset would change with them.

## 8 Conclusion

The published median withdrawal duration for the non-ISO West rests on 12.1 % of the region’s withdrawals, filed with five or six of its thirty-one transmission providers, in service territories covering a minority of the region’s queue. The largest western provider, PacifiCorp, contributes 31 % of the region’s withdrawals and 6 % of its recorded withdrawal dates.

That reporting subset is measurably unlike the region it represents. Its technology mix, queue cohort distribution and interconnection service type all differ at  $p < 10^{-11}$ ; only project capacity does not.

Yet standardising the subset to the region’s actual composition moves the median by less than three months on any specification, and by about three weeks on the one that matches the published estimand—with an interval that includes zero and a sign that reverses under a truncation-matched window. A validation exercise inside the well-covered ISO regions explains why: imposing the West’s compositional distortion on a population whose durations are fully observed shifts the median by roughly eight days. Technology, cohort and capacity account for under 5 % of the variance in withdrawal duration, so a composition gap, even a large and highly significant one, has little room to move a median.

The practical conclusion is not that the published figure is wrong. It is that the caveat usually attached to such figures—that the reporting subset might have an unrepresentative project mix—is the wrong caveat, or at least the minor one. The composition problem is real, statistically unambiguous, and small in effect. The problem that remains is coverage itself: a statistic drawn from one eighth of the population, concentrated in five providers, whose principal risk is that those five may run their queues on a different clock from the twenty-six that publish nothing. That risk cannot be quantified from this data file, and the finding that a quarter of the latent variation in withdrawal risk sits between providers suggests it should not be assumed away.

The most useful thing that could be done with this result is not a better correction. It is a withdrawal date from BPA, Idaho Power, APS, NorthWestern, NV Energy, WAPA or PacifiCorp.

## Reproducing

Requires R (tested on 4.6.1) with `readxl`, `dplyr`, `tidyr`, `ggplot2` and `lme4`. Open `queue-analysis.Rproj` and run in order: `00-setup.R` (shared definitions), `01-explore.R` (the original exploratory session, kept as a record), `02-figures.R`, `03-differences.R`, `04-standardise.R`, `05-logistic.R`, `06-figures.R`.

`04-standardise.R` runs 2000 bootstrap replicates for each of eight estimates and takes several minutes. Both it and `05-logistic.R` set a seed (20260906), so results are exactly reproducible.

## Data and licence

data/LBNL\_Ix\_Queue\_Data\_File\_thru2025.xlsx, *Queued Up* 2026 Edition, from <https://emp.lbl.gov/queues>. Licensed CC BY 4.0; attribution to Lawrence Berkeley National Laboratory and GridTracker. The analysis code in this repository is offered under the same terms.

This paper describes the data file as published and takes no position on the methodology of *Queued Up* itself, whose authors report what transmission providers disclose and are explicit that coverage varies by region.